Testing And Round-Trip Artifacts Guide

This guide explains test execution and round-trip review artifacts.

Run commands from the **UADC_PoC** directory. Test scripts are plain Python scripts and can be executed directly with **$python**. Some can also be run through **pytest** when pytest is installed.

Supporting Files

- **tests/roundtrip/**: reviewable round-trip artifacts. Each case keeps source XML, Structured CSV, xBRL-CSV metadata JSON, and regenerated XML together.
- **tools/build_roundtrip_test_artifacts.py**: rebuilds round-trip artifact sets from sample XML inputs.

Phase 1 Syntax Binding Tests

These tests verify UBL Invoice XML to Structured CSV, xBRL-CSV metadata, and UBL round-trip behavior.

```
& $python .\tests\test_openpeppol_invoice_conversion.py
& $python .\tests\test_lhm_hierarchical_csv_layout.py
& $python .\tests\test_syntax_binding.py
& $python .\tests\test_syntax_binding_reverse.py
& $python .\tests\test_ubl_schema_child_order.py
& $python .\tests\test_bis_billing3_examples_conversion.py
& $python .\tests\test_roundtrip_artifacts.py
```

The forward Phase 1 conversion uses XML parent context recursion, so nested repeated classes preserve parent dimension values such as **dInvoiceLine** with child dimensions such as **dItemAttributes**.

The reverse conversion can derive UBL child element order from XSD files by using **--ubl-schema-root** or **--ubl-schema-url**. **test_ubl_schema_child_order.py** checks this XSD-derived ordering logic without downloading the full UBL package.

test_syntax_binding_reverse.py also checks cross-scope absolute XPath handling: BT-90 must be written below the root **AccountingSupplierParty**, and **PaymentMeans/Invoice** must not exist. **test_bis_billing3_examples_conversion.py** checks the currency-filtered totals in **Allowance-example.xml**: BT-110 is **1225.00** and BT-111 is **9324.00**.

Phase 2 ADS XBRL GL Tests

These tests regenerate Phase 1 Structured CSV first, then use ADS XBRL GL syntax binding CSV files under **specs/bindings/syntax/**.

```
& $python .\tests\test_ads_invoices_received_xbrl_gl.py
& $python .\tests\test_ads_invoices_generated_xbrl_gl.py
& $python .\tests\test_ads_invoices_received_lines_xbrl_gl.py
& $python .\tests\test_ads_invoices_generated_lines_xbrl_gl.py
& $python .\tests\test_ads_supplier_listing_xbrl_gl.py
& $python .\tests\test_ads_customer_master_xbrl_gl.py
```

Expected output:

```
out/phase2/ADS_XBRL_GL/<structured-csv-stem>/<target-view>.xbrl
```

Target view files:

```
Invoices_Received.xbrl
Invoices_Generated.xbrl
Invoices_Received_Lines.xbrl
Invoices_Generated_Lines.xbrl
Supplier_Listing.xbrl
Customer_Master.xbrl
```

test_ads_supplier_listing_xbrl_gl.py verifies that Seller name and identifier are written to the **identifierType=V** identifier reference and that Seller Street, City, Country, and Postal Code are written below its **gl-bus:identifierAddress** tuple.

Phase 2 ADS PSV And CSV Tests

These tests use semantic binding files under **specs/bindings/semantic/**.

```
& $python .\tests\test_ads_invoices_received_psv.py
& $python .\tests\test_ads_invoices_generated_psv.py
& $python .\tests\test_semantic_binding_indexed_paths.py
& $python .\tests\test_semantic_binding_csv_format.py
& $python .\tests\test_phase2_outputs_by_structured_csv_stem.py
```

Expected output:

```
out/phase2/ADS_PSV/<structured-csv-stem>/<target-view>.psv
```

Taxonomy And LHM Checks

```
& $python .\tests\test_lhm_semantic_paths.py
& $python .\tests\test_xbrlgl_generator_uadc_lhm.py
& $python .\tests\validate_taxonomy.py
```

Round-Trip Artifacts

Round-trip flow:

```
source XML -> Structured CSV -> regenerated XML
```

Artifacts are under:

```
tests/roundtrip/<dataset>/
  source_xml/
  structured_csv/
  metadata_json/
  roundtrip_xml/
```

Current datasets:

```
tests/roundtrip/openpeppol-minimal/
tests/roundtrip/bis-billing3-examples/
```

Rebuild:

```
$python = 'python'
& $python .\tools\build_roundtrip_test_artifacts.py
```

Verify:

```
& $python .\tests\test_roundtrip_artifacts.py
```

The test checks source XML, Structured CSV, metadata JSON, and round-trip XML correspondence. It also checks representative values such as **InvoiceNumber**, **DocumentCurrencyCode**, **InvoiceLineIdentifier**, amount **currencyID**, taxonomy entry points, and xBRL-CSV column concept mappings.

Optional Validation

Arelle metadata validation:

```
& $python .\tests\test_xbrl_csv_metadata_arelle.py
```

UBL 2.1 schema validation:

```
& $python .\tests\test_roundtrip_xml_ubl_schema.py
```

The regenerated XML is not expected to be byte-for-byte identical to the source XML. It is reconstructed from bound CSV values and may differ in XML declaration formatting, namespace placement, indentation, and unbound XML content.

Current Test Execution Report

Last recorded report date:

```
2026-07-13
```

Recorded result:

```
PASS
```

The complete set of 22 **tests/test_*.py** regression scripts was executed on 2026-07-13. All 22 completed successfully with no failures.

Scope included EN 16931 LHM-driven syntax binding conversion, Structured CSV generation, xBRL-CSV metadata generation, Arelle validation, UBL reverse conversion, BIS Billing 3 example conversion, LHM checks, and local taxonomy generation checks.

The taxonomy/LHM checks, OpenPeppol conversion, all nine BIS Billing 3 conversions, Structured CSV metadata, ten round-trip artifact cases, and Arelle validation of all ten xBRL-CSV metadata files passed. Absolute currency-filtered XPath evaluation was corrected so **Allowance-example.xml** now writes BT-110 as **1225.00** and BT-111 as **9324.00**. All Phase 2 ADS XBRL GL and ADS PSV/CSV outputs were regenerated and their tests passed.

The Supplier Listing XBRL GL binding was then completed with Seller postal address facts under the **identifierType=V** identifier reference. Supplier Listing was regenerated from all ten Phase 1 inputs, and all ten resulting **Supplier_Listing.xbrl** instances passed Arelle validation. The four ISO 21378 ADC invoice bindings were also applied to all ten Phase 1 inputs, generating 40 CSV target files. Detailed ISO field coverage and known source-data gaps are recorded in **../semantic_binding_conversion/iso21378_adc_invoice_coverage.md**.

These results complete the planned Phase 1 and Phase 2 PoC baseline. ISO 21378 completion here means that the four planned invoice views, their mappings, outputs, and gap classifications are complete; it does not mean that EN 16931 contains every audit-system field defined by ISO 21378.

The reverse converter keeps absolute binding XPaths rooted at the UBL document when a semantic child is stored outside its repeated syntax context. This prevents BT-90 from creating a nested **Invoice** below **PaymentMeans**. All ten regenerated round-trip XML files pass the UBL 2.1 Invoice schema validation when the test is run with an environment containing **xmlschema**.